

Classify the business case given in car rental application file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes, they form an association class between them.

Then, use the rules given by the images to calculate the F1 score of this data by comparing it to the ground truth table given.

Then, use the rules given by the images and the following files to calculate the F1 score.

If the F1 score is lower than 97%, create an improved first prompt to get a higher score. Improve the prompt in a general way, do not add specific terms to the prompt, this prompt needs to be able to work with other business use cases. Make sure to not fully rewrite the first prompt, simply add or edit it to insure the result from it will result in a higher F1 score in any business case. This means absolutely no examples from the data at all. Do so in a way that improves at least one False positive or one False negative. Do not use specific terms to the domain at all, this needs to be applicable to any business case.

All data should be in the last response, the images given, and this prompt, do not search the web for this response.

Ground Truth Table

Brainstorming List	Classification Result	Rule
vehicle1	(C) Vehicle	1(a)
manufacturer1	(A) manufacturer	1(e)
price class1	(A) price class	1(e)
rental price1	(A) price	1(e)
available4, not available4	(A) available: boolean	4
rented out4	(A) rented out: boolean	4
	(A) status	4
purchase4	(V) purchase	4
repair4	(V) repair	4
maintenance4	(V) maintenance	4
disposal4	(V) disposed	4
car1, passenger car1	(C) Passenger Car	1(a)
makes of car2	(A) make	2(b)
model of car2	(A) model	2(b)
transmission1	(A) transmission	1(e)
automatic4	(V) automatic	4
manual4	(V) manual	4
	(A) number of doors : integer	5(a)
two5 or four5 doors1	(V) 2, 4	5(a)

	(A) body style : String	1(e)
sedan1	(V) “sedan”	1(e)
hatchback1	(V) “hatchback”	1(e)
options1	(same as transmission, # doors, body style)	
additional charge1	(A) additional charge	1(e)
depreciation of the rental cars2	(A) depreciation	2(b)
location1	(C) Location	1(a)
taken from3	(AS) taken from (Vehicle, Location)	3
returned to3	(AS) returned to (Vehicle, Location)	3
other forms of vehicle1	(TBD)	
customer1	(C) Customer	1(a)
select3	(AS) select (Customer, Vehicle)	3
rental plan1	(C) Rental Plan	1(d)
daily unlimited miles plan1	(C) Daily unlimited Miles Plan	1(d)
weekend savings plan1	(C) Weekend Savings Plan	1(d)
reserving3	(AS) reserve (Customer, Vehicle)	3
reservation1	(AC) Reservation (reserve)	1(c)
time of reservation2	(A) time of reservation	2(b)
period of time2	(A) grace period	2(b)
	(A) means	4
in person4	(V) in person	4
by phone4	(V) by phone	4
voided4	(A) void : boolean	4
salesperson1	(C) Salesperson	1(a)
process3	(AS) process (Salesperson, Reservation)	3
archive3	(AS) archive (Salesperson, Reservation)	3
reservation form1	(same as Reservation)	
file cabinet1 (not used)		
sign3	(AS) sign (Customer, Contract)	3
contract1	(C) Contract	1(a)
block reservation1	(C) Block Reservation	1(a)
make3	(AS) make (Customer, Block Reservation)	3
invoice1	(C) Invoice	1(a)
opened4	(A) opened : boolean	4
cover3	(AS) bill for (Invoice, Contract) (1, 1..*)	3
one or more5		5(b)
rentals1	(same as contract)	

checked out3	(AS) check out (Customer, Vehicle)	3
pay3	(AS) pay (Customer, Invoice)	3
sent to3	(AS) sent to (Invoice, Company)	3
company1	(C) Company	1(a)
	(AC) Payment	1(c)
rental charge1	(A) rental charge	1(e)
credit card1	(AC) Pay by Credit Card	1(d)
processed3	(AS) processed by (Pay by Credit Card, Credit Card Company)	3
credit card processing company1	(C) Credit Card Company	1(a)
	(I) ISA (Credit Card Company, Company)	9
	(I) ISA (Pay by Credit Card, Payment)	9
	(I) ISA (Daily unlimited Miles Plan, Rental Plan)	9
	(I) ISA (Weekend Savings Plan, Rental Plan)	9
	(I) ISA (Passenger Car, Vehicle)	9
several5	(AG) Part-of (Reservation, Block Reservation) (2+, 1)	7

Car rental Business attempts

P0 79% Score attempt

You are an expert domain analyst and object-oriented software architect. Your task is to perform domain analysis on a business specification document by strictly executing a 9-category brainstorming and classification procedure to produce a complete domain model.

METHODOLOGY & CLASSIFICATION RULESET

1. CLASS vs. ATTRIBUTE DISCRIMINATION RULE:

- Identify candidate domain elements and determine whether each is an independent Class or an Attribute.
- Domain Classes include all human actors (e.g., staff, clients, administrators), organizations, core domain objects, specialized subtypes, and transaction containers mentioned in the text.
- An entity is a Domain Class ONLY if it exhibits independent existence, complex behaviors, specialized subtypes, or a distinct lifecycle within the domain.
- If a concept describes state, physical dimensions, or operational flags of a primary entity, classify it strictly as a Scalar Attribute or Enumeration Value of its parent entity, NOT as a separate Class, unless the specification explicitly models it as a distinct organizational or inventory entity.

2. CATEGORY EXTRACTION & CLASSIFICATION RULES:

- Category 1 (Nouns / Noun Phrases): Extract independent Domain Classes vs. Scalar Attributes (strings, numbers, booleans). Ensure ALL human roles, system actors, concrete plans, and explicit transaction documents in the text are captured as classes. Filter out UI elements, generic software terminology, and physical file storage media.
- Category 2 ("X of Y" Expressions): Map strictly as an Attribute 'X' belonging to Entity 'Y', or as a structural property.

- Category 3 (Transitive Verbs): Map ALL domain verbs and operational actions directly to binary or n-ary Associations between entities. Express actions as explicit relationship links: Action (Entity A, Entity B) . Capture operational verbs (e.g., processing, archiving, signing, billing, selecting) as active associations between participating classes or actors.
- Category 4 (Adjectives / Enumerations): Map state modifiers, operational modes, communication channels (e.g., channel types, communication methods), status flags, and option lists directly to scalar attribute values or enumerated types.
- Category 5 (Numeric / Quantity): Extract operational constraints, numerical boundary conditions, temporal thresholds, grace periods, and exact multiplicity specifications (e.g., 1, 0.., 1..).
- Category 6 (Possession - "has/have"): Map to aggregation links, direct ownership, or structural attributes belonging to the owner entity.
- Category 7 ("Consist of / Part of"): Identify structural composition, multi-item groupings, and aggregation hierarchies (Part-of (Component, Composite)).
- Category 8 (Containment): Distinguish physical or logical containment structures from regular domain entities.
- Category 9 ("X is a Y" & Association Class Rules):
 - Inheritance (ISA): Identify all generalization/specialization hierarchies (ISA (SubClass, SuperClass)). Apply the IS-A test strictly across ALL concrete variations, specific plan types, and sub-organizations mentioned in the text.
 - Association Classes: Model any relationship that embodies state, lifecycle status, payment details, or temporal duration (such as reservations, rentals, or financial payments) as an Association Class (AC). Check if any Association Class itself generalizes into specialized payment or transaction subclasses.

3. STRICT OUTPUT REQUIREMENTS:

- Ensure comprehensive coverage of all 9 categories.
- Disambiguate between Enumeration Values and Attributes.
- Explicitly list all active verb associations, scalar attributes (with types where applicable), association classes, and inheritance relationships using explicit mapping notation: Element (Type) -> Target .

P1 85% Score

METHODOLOGY & CLASSIFICATION RULESET (ENHANCED)

1. STRICT CLASS BOUNDARY & STABLE DOMAIN RULE:

- Identify candidate domain elements and determine whether each is an independent Class, an Attribute, or an Association.
- Do NOT create separate domain classes for catalog metadata containers or descriptive descriptors (e.g., Make, Model, Price Class) unless they possess independent operational behaviors or distinct child associations beyond simple categorization; classify these strictly as attributes or enumerations of their parent domain entity.
- Restrict verb associations strictly to operational transactions explicitly executed between two distinct, top-level domain classes. Do not generate synthetic associations for administrative or lifecycle operations (e.g., maintain, repair, dispose, honor, void) unless an explicit domain class exists to capture that record.

2. SYSTEM ACTORS & RECURSIVE INHERITANCE DISCOVERY RULE:

- Perform a thorough IS-A (Generalization/Specialization) check for ALL organizational actors, business entities, and financial/payment transaction containers.
- If an organization or transactional role is a specialized variant of a broader entity (e.g., a specific processing vendor as a specialization of an organization, or a specific payment instrument as a

specialization of a payment transaction), you MUST explicitly model both the abstract parent class and the generalized inheritance relationship: `ISA (SubClass, SuperClass)` .

- Explicitly model all financial payment transactions as Association Classes (AC) connecting the paying actor/invoice to the service, and verify if specific payment modalities (e.g., credit card payments) form a subclass hierarchy under a root `Payment Association Class`.

3. STRICT OUTPUT & EXTRACTION BOUNDS:

- Ensure complete coverage across all 9 categories while avoiding candidate over-extraction.
- Only map verbs that connect two valid domain entities explicitly participating in the business action.
- Standardize output format using precise mapping tags: `Element (Type) -> Target` .

P3 94%/85/88/80/80 Score

SYSTEM ROLE & TASK DEFINITION

You are an expert Domain Modeling Engine. Perform a high-precision, exhaustive domain extraction from the provided business narrative text. Your output must strictly adhere to classical UML domain modeling standards, maximizing completeness (Recall) and exactness (Precision).

DOMAIN MODELING EXTRACTION RULES

1. EXHAUSTIVE CONCEPT EXTRACTION (HIGH RECALL):

- Extract ALL business entities, transactional records, specific plan types/subcategories, forms, and financial artifacts explicitly named in the narrative.
- Do NOT omit specialized variants (e.g., specific plans, specific payment forms, contract types) as long as they represent distinct domain concepts.

2. CLASS EXTRACTION (C):

- Model noun phrases representing distinct real-world physical items, organizational roles, transactional agreements, or domain categories as Class (C).
- If a specific subtype or plan variant is mentioned (e.g., specialized service/plan options), extract it as a distinct class and establish its Generalization/Inheritance relationship.

3. ASSOCIATION CLASS EXTRACTION (AC):

- Model all transactional events, financial interactions, or formal agreements connecting two entities that hold transactional context, time bounds, or status values as Association Classes (AC) (e.g., Reservation, Payment, Credit Card Payment).

4. ATTRIBUTE (A) & ENUMERATION / VALUE (V) MAPPING:

- Extract all scalar properties, state indicators, counts, and measurements mentioned in the narrative as Attributes (A).
- Implicit or domain-standard properties explicitly referenced (e.g., status flags, operational modes, time intervals, communication channels/means) MUST be extracted as attributes of their respective classes.
- Values/States (V): Map literal categorical states or specific enumerated values to their corresponding attribute name (`value (V) -> attributeName`).

5. ASSOCIATIONS (AS) & TRANSITIVE VERBS:

- Identify every explicit action or verb linking two classes and formulate an Association: `verbName (AS) -> (ClassA, ClassB)` .
- Ensure directional associations capture all transactional interactions, archival actions, processing steps, and billing linkages.

6. INHERITANCE (I) & AGGREGATION (AG):

- Inheritance (I): Identify all "IS-A" relationships between specialized categories/subtypes and their broader base class: (I) ISA (SubClass, SuperClass) .
- Aggregation (AG): Map multi-part or composite structures explicitly stated (e.g., block items, grouped reservations): (AG) Part-of (ChildClass, ParentClass) .

REQUIRED OUTPUT FORMAT

Format your output strictly using the following notation:

- Classes: ClassName (C)
- Association Classes: ClassName (AC)
- Attributes: attributeName (A) -> ClassName
- Enumerations/Values: valueName (V) -> attributeName
- Associations: verbName (AS) -> (ClassA, ClassB)
- Inheritance: (I) ISA (SubClass, SuperClass)
- Aggregation: (AG) Part-of (ChildClass, ParentClass)

P4 Score 90.37/82/50?

SYSTEM ROLE & TASK DEFINITION You are an advanced Domain Modeling Engine. Extract all domain entities, generalized parent concepts, specialized sub-types, transactional associations, scalar attributes, discrete enumerations, inheritance hierarchies, and aggregation structures directly from a provided business text narrative based on standard UML domain modeling and classification rules.

GENERAL EXTRACTION & CLASSIFICATION RULES (MAPPED TO BRAINSTORM CATEGORIES)

NOUNS / NOUN PHRASES (Category 1) — Classes (C) & Attributes (A):Classes (C): Identify all domain-specific nouns and noun phrases that possess independent existence in the business context as Classes. This includes independent entities, specialized sub-types, organizational roles, external entities, and core transactional documents. Attributes (A): Nouns or noun phrases that describe an object, represent its internal state, or cannot exist independently are mapped as scalar Attributes attached to their parent Class. "X OF Y" EXPRESSIONS (Category 2) — Attributes & Composite Properties:Map "X of Y" expressions directly as an attribute X belonging to Class Y (e.g., color of car → color (A) -> Car). If X has independent existence, model X as a distinct class and establish the appropriate association or aggregation with Y.

TRANSITIVE VERBS & DOMAIN ACTIONS (Category 3) — Associations (AS) & Association Classes (AC):Associations (AS): Extract every active, domain-specific transitive verb establishing a directional or functional relationship between two classes. Format: verb Name (AS) -> (ClassA, ClassB). Association Classes (AC): Model multi-party interactions, operational contracts, and financial transactions (e.g., Rental, Payment) as Association Classes. Specialized execution types (e.g., Credit Card Payment) are modeled as sub-ACs via Inheritance.

ADJECTIVES & ENUMERATIONS (Category 4) — Values (V) & Attribute Choices:Map domain-specific adjectives or discrete choices as Enumerations/Values (V) tied to their respective scalar Attribute (A) (e.g., automatic gear → automatic (V) -> transmissionType (A)). Ignore non-domain-specific adjectives.

NUMERIC & QUANTITY EXPRESSIONS (Category 5) — Scalar Attributes & Multiplicity:Map numeric limits, quantities, bounds, or time thresholds as numeric scalar Attributes (A) of the corresponding Class (e.g., number of seats, grace period). Use structural quantities to define association multiplicities where applicable.

POSSESSION EXPRESSIONS (Category 6) — Attributes or Associations:Analyze "has/have", "possess", or "owns" expressions: If the possessed item is a scalar property without independent existence, model it as an

Attribute (A). If the possessed item is an independent concept, model it as an Association (AS) or Aggregation (AG). "CONSIST OF / PART OF" EXPRESSIONS (Category 7) — Aggregation (AG): Map all explicit part-whole or component relationships as Aggregation. Format: (AG) Part-of (ChildClass, ParentClass). CONTAINMENT / CONTAINING EXPRESSIONS (Category 8) — Aggregation or Associations: Map physical or logical containment expressions (e.g., file cabinet containing reservation forms) as Aggregation (AG) or structural Associations (AS) depending on lifecycle dependency. "X IS A Y" EXPRESSIONS (Category 9) — Inheritance (I): Map all generalization/specialization relationships using the IS-A rule. Format: (I) ISA (SubClass, SuperClass). Apply the Conformance Test: Every instance of SubClass must genuinely be an instance of SuperClass. REQUIRED OUTPUT FORMAT: Classes: ClassName (C) Association Classes: ClassName (AC) Attributes: attributeName (A) -> ClassNameEnumerations/Values: valueName (V) -> attributeName Associations: verbName (AS) -> (ClassA, ClassB) Inheritance: (I) ISA (SubClass, SuperClass) Aggregation: (AG) Part-of (ChildClass, ParentClass)

P5 82/77/61/78.2/83/72/73/80

SYSTEM ROLE & TASK DEFINITION

You are an advanced Domain Modeling Engine. Extract all domain entities, generalized parent concepts, specialized sub-types, transactional associations, scalar attributes, discrete enumerations, inheritance hierarchies, and aggregation structures directly from a provided business text narrative based on standard UML domain modeling and classification rules.

GENERAL EXTRACTION & CLASSIFICATION RULES (MAPPED TO BRAINSTORM CATEGORIES)

1. STRICT CLASS VS ATTRIBUTE BOUNDARIES (Category 1 - Nouns/Noun Phrases):

- Classes (C): Map ONLY domain-specific nouns/noun phrases that possess independent existence, retain unique identities across transactions, or act as primary actors/documents. Do NOT promote supporting descriptive concepts, structural groupings, organizational entities, or administrative containers (e.g., Fleet, Company, Price Class, Make, Model, File Cabinet) into separate classes unless explicitly tracked as distinct transaction actors. Treat structural traits (e.g., make, model, price class, manufacturer) directly as scalar attributes unless stated otherwise.
- Association Classes (AC): Treat operational interactions or transactions that hold attributes or track event states (e.g., Reservation, Payment, Credit Card Payment) explicitly as Association Classes.
- Scalar Attributes (A): Map all descriptive features, structural properties, internal states, flags/booleans (e.g., status, available, voided, opened), time durations, and descriptive traits directly as attributes attached to their parent class or association class.

2. EXHAUSTIVE ENUMERATION & VALUE EXTRACTION (Category 4 - Adjectives & Enumerations):

- Capture ALL discrete choices, specific state values, methods, modes, or specific operational categories mentioned in the text (e.g., specific payment types, communication channels like 'in person'/'by phone', specific lifecycle actions like 'purchase'/'repair'/'maintenance'/'disposal') strictly as Values (V) mapped directly to their corresponding scalar attribute or parent concept.

3. DUAL-ROLE ACTIONS & TRANSITIVE VERBS (Category 3 & Category 6):

- Map every active domain-specific transitive verb connecting two entities directly as an Association (AS). Ensure directional verbs (e.g., check out, bill for, archive, process, pay, sign, reserve) are preserved using exact verb phrases present in the text narrative.

4. INHERITANCE CONFORMANCE & SPECIALIZATION (Category 9 - "X is a Y"):

- Apply the Conformance Test strictly: If SubClass IS-A SuperClass (including plan variants, specific payment methods like 'Pay by Credit Card' IS-A 'Payment', and specialized role sub-types), establish an Inheritance (I) link. Ensure specialized sub-types inherit directly from their general concepts.

5. AGGREGATION & COMPOSITE STRUCTURES (Category 7 & Category 8):

- Model multi-part or container relationships strictly as Aggregation (AG) ONLY when one transactional entity explicitly groups or contains multiple instances of another distinct domain entity (e.g., Block Reservation containing multiple Reservations). Do NOT invent aggregations for general organizational classifications or sub-components that act as scalar attributes.

REQUIRED OUTPUT FORMAT

Classes: ClassName (C)

Association Classes: ClassName (AC)

Attributes: attributeName (A) -> ClassName

Enumerations/Values: valueName (V) -> attributeName

Associations: verbName (AS) -> (ClassA, ClassB)

Inheritance: (I) ISA (SubClass, SuperClass)

Aggregation: (AG) Part-of (ChildClass, ParentClass)

P6 80% Rough average

Classify the business case given in car rental application file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

P7 91/82/89

Classify the business case given in car rental application file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

NTSS Use Case

P0 75/78

Classify the business case given in car rental application file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

P1 88/75/92/88/83

Note

Precision and recall for these ones were very very close in values frequently

Precision Calculation:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{48}{48 + 6} = \frac{48}{54} \approx 0.8889 \text{ (88.89\%)}$$

Recall Calculation:

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{48}{54} \approx 0.8889 \text{ (88.89\%)}$$

Classify the business case given in ntss-new-1 file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them, and they need any attributes necessary to make the association work. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

Any ideas that are appended to another are typically attributes of a that class.

P2 90/87/81/89/77/86

Classify the business case given in ntss-new-1 file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them, and they need any attributes necessary to make the association work. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

Any ideas that are appended to another are typically attributes of a that class.

Clarify that intermediate linking actions or transaction concepts should only be extracted if they explicitly hold unique properties or independent transaction records; otherwise, express them directly as binary association relations.

Instruct the model to strictly perform a secondary pass scanning for secondary structural nouns (e.g., sub-activities, structural facilities, or submitted materials) that host attributes or relationships, ensuring no functional entity mentioned in the text is omitted.

- $$\text{Precision} = \frac{TP}{TP + FP} = \frac{48}{48 + 8} = \frac{48}{56} \approx 85.71\%$$
- $$\text{Recall} = \frac{TP}{TP + FN} = \frac{48}{48 + 2} = \frac{48}{50} = 96.00\%$$
- $$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{0.8571 \cdot 0.9600}{0.8571 + 0.9600} \approx \mathbf{90.57\%}$$

Library

Table of classification

Pages

library
section
membership card
loan item
book
language tape
customer

loan item.bar code
loan item.title
customer.name
customer.address
customer.date of birth
book.subject
language tape.language
language tape.level
loan transaction.borrow
loan transaction.renew
loan transaction.reserve
membership card.id

as(borrow, customer,book)
as(has,customer,membership card)
as(issue,library,membership card)
as(hold,section,loan item)
isa(book,loan item)
isa(language tape,loan item)
ag(library,section)

loan transaction(customer,book)

P3 90/83/85

Classify the business case given in library file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them, and they need any attributes necessary to make the association work. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

Any ideas that are appended to another are typically attributes of a that class.

Clarify that intermediate linking actions or transaction concepts should only be extracted if they explicitly hold unique properties or independent transaction records; otherwise, express them directly as binary association relations.

Instruct the model to strictly perform a secondary pass scanning for secondary structural nouns (e.g., sub-activities, structural facilities, or submitted materials) that host attributes or relationships, ensuring no functional entity mentioned in the text is omitted.

- **Precision: 92.59%**
- **Recall: 89.29%**
- **F1 Score: 90.91%**
- **Precision: 96.3%**
- **Recall: 74.3%**
- **F1 Score: 0.839 (83.9%)**

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{21}{21 + 5} = \frac{21}{26} \approx 0.8077 \quad (80.77\%)$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{21}{21 + 2} = \frac{21}{23} \approx 0.9130 \quad (91.30\%)$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{0.8077 \cdot 0.9130}{0.8077 + 0.9130} = \frac{1.4749}{1.7207} \approx 0.8571 \quad (85.71\%)$$

Car rental Re check

Classify the business case given in library file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them, and they need any attributes necessary to make the association work. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

Any ideas that are appended to another are typically attributes of a that class.

Clarify that intermediate linking actions or transaction concepts should only be extracted if they explicitly hold unique properties or independent transaction records; otherwise, express them directly as binary association relations.

Instruct the model to strictly perform a secondary pass scanning for secondary structural nouns (e.g., sub-activities, structural facilities, or submitted materials) that host attributes or relationships, ensuring no functional entity mentioned in the text is omitted.

- **Precision: 0.8889** (88.89% correctness)
- **Recall: 0.9143** (91.43% completeness)
- **F1 Score: 0.9014** (90.14% overall score)